

bayesian networks

CSCI
270

$$P(x_f, x_{\sigma}, x, y_f, y_{\sigma}, y, z_f, z_{\sigma}, z)$$



$$= P(z \mid x_f, x_{\sigma}, x, y_f, y_{\sigma}, y, z_f, z_{\sigma})$$

$$\cdot P(z_{\sigma} \mid x_f, x_{\sigma}, x, y_f, y_{\sigma}, y, z_f)$$

$$\cdot P(z_f \mid x_f, x_{\sigma}, x, y_f, y_{\sigma}, y)$$

$$\cdot P(y \mid x_f, x_{\sigma}, x, y_f, y_{\sigma})$$

$$\cdot P(y_{\sigma} \mid x_f, x_{\sigma}, x, y_f)$$

$$\cdot P(y_f \mid x_f, x_{\sigma}, x)$$

$$\cdot P(x \mid x_f, x_{\sigma})$$

$$\cdot P(x_{\sigma} \mid x_f)$$

$$\cdot P(x_f)$$

can we express this
more compactly?

	x_f	x_{σ}	x	y_f	y_{σ}	y	z_f	z_{σ}	z	
joint distribution	A	A	A	A	A	A	A	A	A	p_1
	A	A	A	A	A	A	A	A	B	p_2
										$\vdots$
										p_{46656}
	0	0	0	0	0	0	0	0	0	

46656 rows



$$P(x_{\phi}, x_{\sigma}, x, y_{\phi}, y_{\sigma}, y, z_{\phi}, z_{\sigma}, z) \leftarrow 46655 \text{ values}$$



$$= P(z \mid x_{\phi}, x_{\sigma}, x, y_{\phi}, y_{\sigma}, y, z_{\phi}, z_{\sigma})$$

$$\cdot P(z_{\sigma} \mid x_{\phi}, x_{\sigma}, x, y_{\phi}, y_{\sigma}, y, z_{\phi})$$

$$\cdot P(z_{\phi} \mid x_{\phi}, x_{\sigma}, x, y_{\phi}, y_{\sigma}, y)$$

$$\cdot P(y \mid x_{\phi}, x_{\sigma}, x, y_{\phi}, y_{\sigma})$$

$$\cdot P(y_{\sigma} \mid x_{\phi}, x_{\sigma}, x, y_{\phi})$$

$$\cdot P(y_{\phi} \mid x_{\phi}, x_{\sigma}, x)$$

$$\cdot P(x \mid x_{\phi}, x_{\sigma})$$

$$\cdot P(x_{\sigma} \mid x_{\phi})$$

$$\cdot P(x_{\phi})$$

46655 values



$$P(x_{\phi}, x_{\sigma}, x, y_{\phi}, y_{\sigma}, y, z_{\phi}, z_{\sigma}, z)$$



$$= P(z \mid x_{\phi}, x_{\sigma}, x, y_{\phi}, y_{\sigma}, y, z_{\phi}, z_{\sigma})$$

$$\cdot P(z_{\sigma} \mid x_{\phi}, x_{\sigma}, x, y_{\phi}, y_{\sigma}, y, z_{\phi})$$

$$\cdot P(z_{\phi} \mid x_{\phi}, x_{\sigma}, x, y_{\phi}, y_{\sigma}, y)$$

$$\cdot P(y \mid x_{\phi}, x_{\sigma}, x, y_{\phi}, y_{\sigma})$$

$$\cdot P(y_{\sigma} \mid x_{\phi}, x_{\sigma}, x, y_{\phi})$$

$$\cdot P(y_{\phi} \mid x_{\phi}, x_{\sigma}, x)$$

$$\cdot P(x \mid x_{\phi}, x_{\sigma})$$

$$\cdot P(x_{\sigma} \mid x_{\phi})$$

$$\cdot P(x_{\phi})$$

but we can take
advantage of
conditional
independence

$$P(x_{\phi}, x_{\sigma}, x, y_{\phi}, y_{\sigma}, y, z_{\phi}, z_{\sigma}, z)$$



$$= P(z \mid x_{\phi}, x_{\sigma}, x, y_{\phi}, y_{\sigma}, y, z_{\phi}, z_{\sigma})$$

$$\cdot P(z_{\sigma} \mid x_{\phi}, x_{\sigma}, x, y_{\phi}, y_{\sigma}, y, z_{\phi})$$

$$\cdot P(z_{\phi} \mid x_{\phi}, x_{\sigma}, x, y_{\phi}, y_{\sigma}, y)$$

$$\cdot P(y \mid x_{\phi}, x_{\sigma}, x, y_{\phi}, y_{\sigma})$$

$$\cdot P(y_{\sigma} \mid x_{\phi}, x_{\sigma}, x, y_{\phi})$$

$$\cdot P(y_{\phi} \mid x_{\phi}, x_{\sigma}, x)$$

$$\cdot P(x \mid x_{\phi}, x_{\sigma})$$

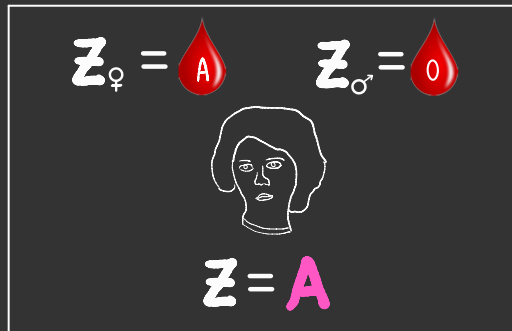
$$\cdot P(x_{\sigma} \mid x_{\phi})$$

$$\cdot P(x_{\phi})$$

observe:

if we know z_{ϕ} and z_{σ} , then

the remaining variables provide no additional information about z



So:

$$z \perp\!\!\!\perp \{x_{\phi}, x_{\sigma}, x, y_{\phi}, y_{\sigma}, y\} \mid \{z_{\phi}, z_{\sigma}\}$$

$$P(x_{\text{♀}}, x_{\text{♂}}, x, y_{\text{♀}}, y_{\text{♂}}, y, z_{\text{♀}}, z_{\text{♂}}, z)$$



$$= P(z \mid x_{\text{♀}}, x_{\text{♂}}, x, y_{\text{♀}}, y_{\text{♂}}, y, z_{\text{♀}}, z_{\text{♂}})$$

$$\cdot P(z_{\text{♂}} \mid x_{\text{♀}}, x_{\text{♂}}, x, y_{\text{♀}}, y_{\text{♂}}, y, z_{\text{♀}})$$

$$\cdot P(z_{\text{♀}} \mid x_{\text{♀}}, x_{\text{♂}}, x, y_{\text{♀}}, y_{\text{♂}}, y)$$

$$\cdot P(y \mid x_{\text{♀}}, x_{\text{♂}}, x, y_{\text{♀}}, y_{\text{♂}})$$

$$\cdot P(y_{\text{♂}} \mid x_{\text{♀}}, x_{\text{♂}}, x, y_{\text{♀}})$$

$$\cdot P(y_{\text{♀}} \mid x_{\text{♀}}, x_{\text{♂}}, x)$$

$$\cdot P(x \mid x_{\text{♀}}, x_{\text{♂}})$$

$$\cdot P(x_{\text{♂}} \mid x_{\text{♀}})$$

$$\cdot P(x_{\text{♀}})$$



Xena

X

Xena's bloodtype

X_♀

gene that Xena inherited from her mother

X_♂

gene that Xena inherited from her father



Yves

Y

Yves's bloodtype

Y_♀

gene that Yves inherited from his mother

Y_♂

gene that Yves inherited from his father



Zelda

Z

Zelda's bloodtype

Z_♀

gene that Zelda inherited from her mother

Z_♂

gene that Zelda inherited from her father

what other
conditional independence
relationships are there?

$$P(x_{\text{♀}}, x_{\text{♂}}, x, y_{\text{♀}}, y_{\text{♂}}, y, z_{\text{♀}}, z_{\text{♂}}, z)$$



$$= P(z \mid x_{\text{♀}}, x_{\text{♂}}, x, y_{\text{♀}}, y_{\text{♂}}, y, z_{\text{♀}}, z_{\text{♂}})$$

$$\cdot P(z_{\text{♂}} \mid x_{\text{♀}}, x_{\text{♂}}, x, y_{\text{♀}}, y_{\text{♂}}, y, z_{\text{♀}})$$

$$\cdot P(z_{\text{♀}} \mid x_{\text{♀}}, x_{\text{♂}}, x, y_{\text{♀}}, y_{\text{♂}}, y)$$

$$\cdot P(y \mid x_{\text{♀}}, x_{\text{♂}}, x, y_{\text{♀}}, y_{\text{♂}})$$

$$\cdot P(y_{\text{♂}} \mid x_{\text{♀}}, x_{\text{♂}}, x, y_{\text{♀}})$$

$$\cdot P(y_{\text{♀}} \mid x_{\text{♀}}, x_{\text{♂}}, x)$$

$$\cdot P(x \mid x_{\text{♀}}, x_{\text{♂}})$$

$$\cdot P(x_{\text{♂}} \mid x_{\text{♀}})$$

$$\cdot P(x_{\text{♀}})$$



Xena

X Xena's bloodtype

X_♀ gene that Xena inherited from her mother

X_♂ gene that Xena inherited from her father



Yves

Y Yves's bloodtype

Y_♀ gene that Yves inherited from his mother

Y_♂ gene that Yves inherited from his father



Zelda

Z Zelda's bloodtype

Z_♀ gene that Zelda inherited from her mother

Z_♂ gene that Zelda inherited from her father

what other
conditional independence
relationships are there?

$P(x_{\phi}, x_{\sigma}, x, y_{\phi}, y_{\sigma}, y, z_{\phi}, z_{\sigma}, z) \leftarrow 46656 \text{ rows}$



$= P(z \mid z_{\phi}, z_{\sigma}) \leftarrow 36 \text{ rows}$

$\cdot P(z_{\sigma} \mid x_{\phi}, x_{\sigma}) \leftarrow 27 \text{ rows}$

$\cdot P(z_{\phi} \mid y_{\phi}, y_{\sigma}) \leftarrow 27 \text{ rows}$

$\cdot P(y \mid y_{\phi}, y_{\sigma}) \leftarrow 36 \text{ rows}$

$\cdot P(y_{\sigma}) \leftarrow 3 \text{ rows}$

$\cdot P(y_{\phi}) \leftarrow 3 \text{ rows}$

$\cdot P(x \mid x_{\phi}, x_{\sigma}) \leftarrow 36 \text{ rows}$

$\cdot P(x_{\sigma}) \leftarrow 3 \text{ rows}$

$\cdot P(x_{\phi}) \leftarrow 3 \text{ rows}$

174 rows

yes

can we express this more compactly?

	x_{σ}	x_{ϕ}	x	y_{σ}	y_{ϕ}	y	z_{σ}	z_{ϕ}	z	
joint distribution	A	A	A	A	A	A	A	A	A	P_1
	A	A	A	A	A	A	A	A	B	P_2
	:	:	:	:	:	:	:	:	:	$\vdots$
	A	O	O	A	O	O	A	O	O	P_{max}

46656 rows



$$P(x_{\phi}, x_{\sigma}, x, y_{\phi}, y_{\sigma}, y, z_{\phi}, z_{\sigma}, z)$$



$$= P(z \mid z_{\phi}, z_{\sigma})$$

$$\cdot P(z_{\sigma} \mid x_{\phi}, x_{\sigma})$$

$$\cdot P(z_{\phi} \mid y_{\phi}, y_{\sigma})$$

$$\cdot P(y \mid y_{\phi}, y_{\sigma})$$

$$\cdot P(y_{\sigma})$$

$$\cdot P(y_{\phi})$$

$$\cdot P(x \mid x_{\phi}, x_{\sigma})$$

$$\cdot P(x_{\sigma})$$

$$\cdot P(x_{\phi})$$

we can visualize the
independence relationships
using a **directed graph**

create a **vertex** for each variable

$$P(x_{\text{♀}}, x_{\text{♂}}, x, y_{\text{♀}}, y_{\text{♂}}, y, z_{\text{♀}}, z_{\text{♂}}, z)$$



$$= P(z \mid z_{\text{♀}}, z_{\text{♂}})$$

$$\cdot P(z_{\text{♂}} \mid x_{\text{♀}}, x_{\text{♂}})$$

$$\cdot P(z_{\text{♀}} \mid y_{\text{♀}}, y_{\text{♂}})$$

$$\cdot P(y \mid y_{\text{♀}}, y_{\text{♂}})$$

$$\cdot P(y_{\text{♂}})$$

$$\cdot P(y_{\text{♀}})$$

$$\cdot P(x \mid x_{\text{♀}}, x_{\text{♂}})$$

$$\cdot P(x_{\text{♂}})$$

$$\cdot P(x_{\text{♀}})$$

X

Y

X_♀

X_♂

Y_♀

Y_♂

Z_♀

Z_♂

Z

the conditional set of a **variable**
are its **parents** in the graph

$$P(x_{\phi}, x_{\sigma}, x, y_{\phi}, y_{\sigma}, y, z_{\phi}, z_{\sigma}, z)$$



$$= P(z \mid z_{\phi}, z_{\sigma})$$

$$\cdot P(z_{\sigma} \mid x_{\phi}, x_{\sigma})$$

$$\cdot P(z_{\phi} \mid y_{\phi}, y_{\sigma})$$

$$\cdot P(y \mid y_{\phi}, y_{\sigma})$$

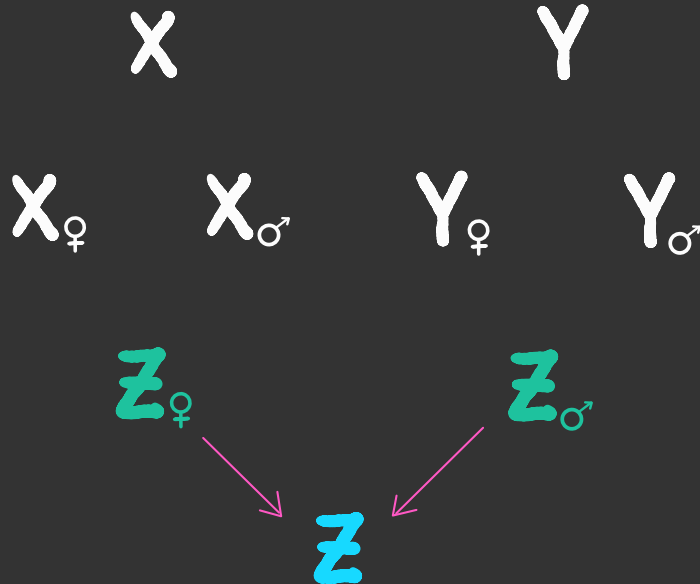
$$\cdot P(y_{\sigma})$$

$$\cdot P(y_{\phi})$$

$$\cdot P(x \mid x_{\phi}, x_{\sigma})$$

$$\cdot P(x_{\sigma})$$

$$\cdot P(x_{\phi})$$



this graph is called a **bayesian network**

$$P(x_{\phi}, x_{\sigma}, x, y_{\phi}, y_{\sigma}, y, z_{\phi}, z_{\sigma}, z)$$



$$= P(z \mid z_{\phi}, z_{\sigma})$$

$$\cdot P(z_{\sigma} \mid x_{\phi}, x_{\sigma})$$

$$\cdot P(z_{\phi} \mid y_{\phi}, y_{\sigma})$$

$$\cdot P(y \mid y_{\phi}, y_{\sigma})$$

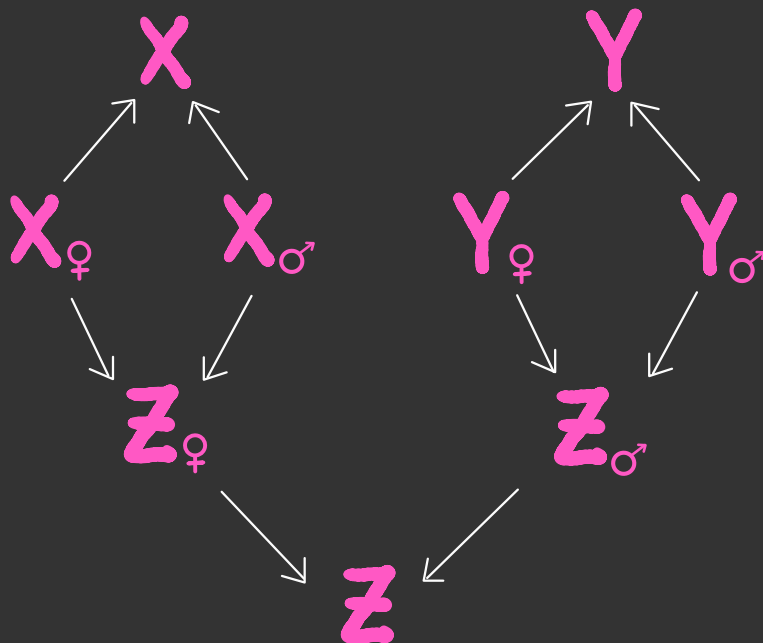
$$\cdot P(y_{\sigma})$$

$$\cdot P(y_{\phi})$$

$$\cdot P(x \mid x_{\phi}, x_{\sigma})$$

$$\cdot P(x_{\sigma})$$

$$\cdot P(x_{\phi})$$



this graph is called a **bayesian network**

$$P(x_{\phi}, x_{\sigma}, x, y_{\phi}, y_{\sigma}, y, z_{\phi}, z_{\sigma}, z)$$



$$= P(z \mid z_{\phi}, z_{\sigma})$$

$$\cdot P(z_{\sigma} \mid x_{\phi}, x_{\sigma})$$

$$\cdot P(z_{\phi} \mid y_{\phi}, y_{\sigma})$$

$$\cdot P(y \mid y_{\phi}, y_{\sigma})$$

$$\cdot P(y_{\sigma})$$

$$\cdot P(y_{\phi})$$

$$\cdot P(x \mid x_{\phi}, x_{\sigma})$$

$$\cdot P(x_{\sigma})$$

$$\cdot P(x_{\phi})$$

